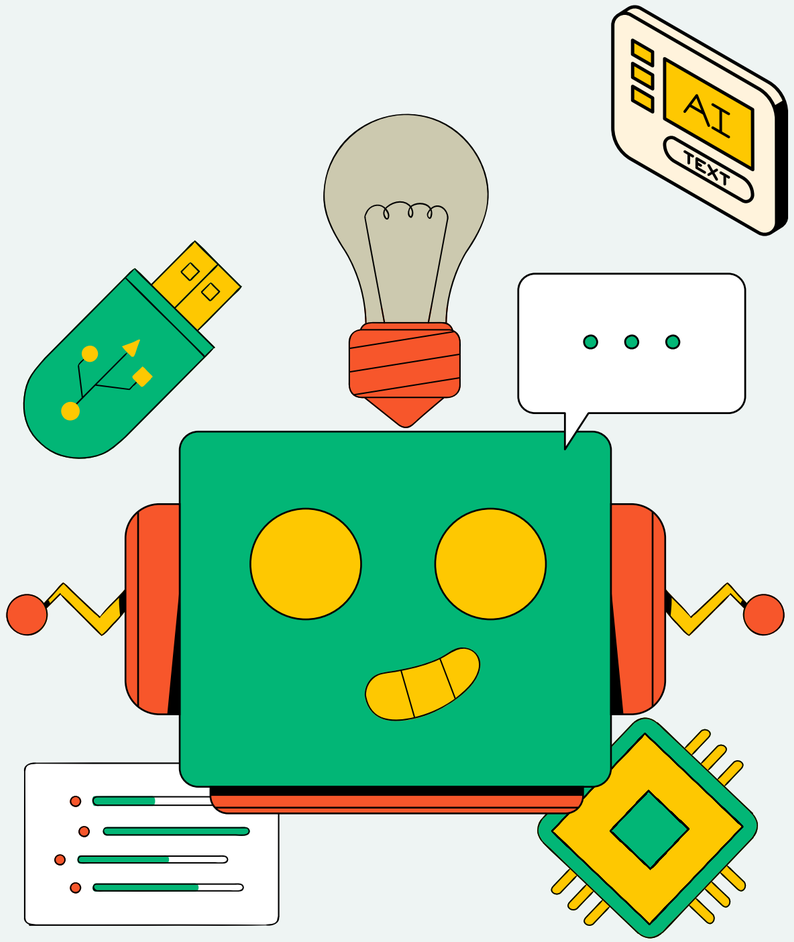
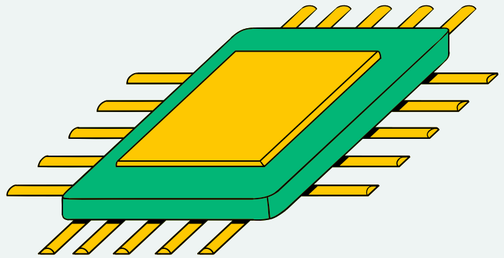


AI application for Singapore Building Regulatory Approval Process

PRESENTED BY: MAN TING

INTERNSHIP DURATION: SEP24–JAN25



BACKGROUND INFORMATION

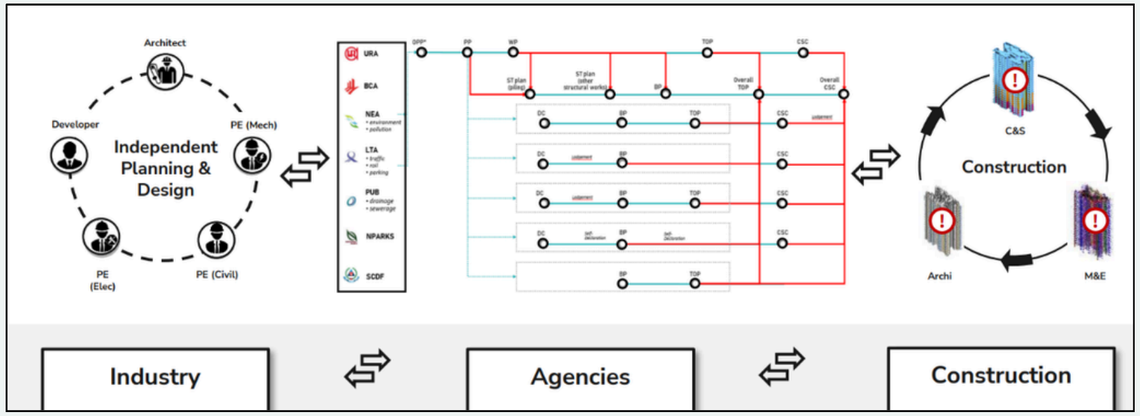
Corenet X (CX) is a multi-agency regulatory submission gateway for building projects. A total of 8 government agencies were involved in the CX project which includes: URA BCA NEA NParks LTA PUB SCDF and HDB

It changes the current practice of QPs* dealing separately with multiple regulatory agencies, and producing different versions of building plans thereafter.

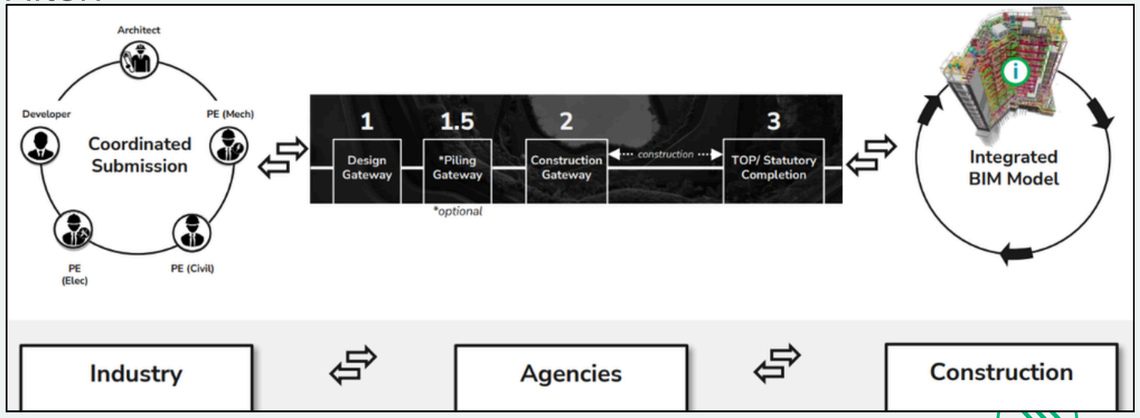
It enables project teams to:

- Collaboratively review designs
- Detect conflicts
- Create a coordinated OpenBIM* model for regulatory approval

Before:

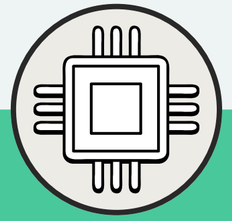


After:



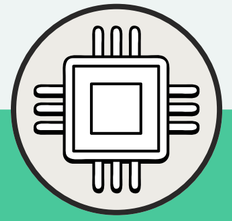
*Qualified Person (e.g. professional engineers and registered architects)
*Open Building Information Modelling (OpenBIM)

THIS PROCESS LEADS TO A FEW INCONVENIENCE:



Time-Consuming

Manually verifying compliance could be time-consuming, especially for complex building with numerous elements.



Resource Intensive

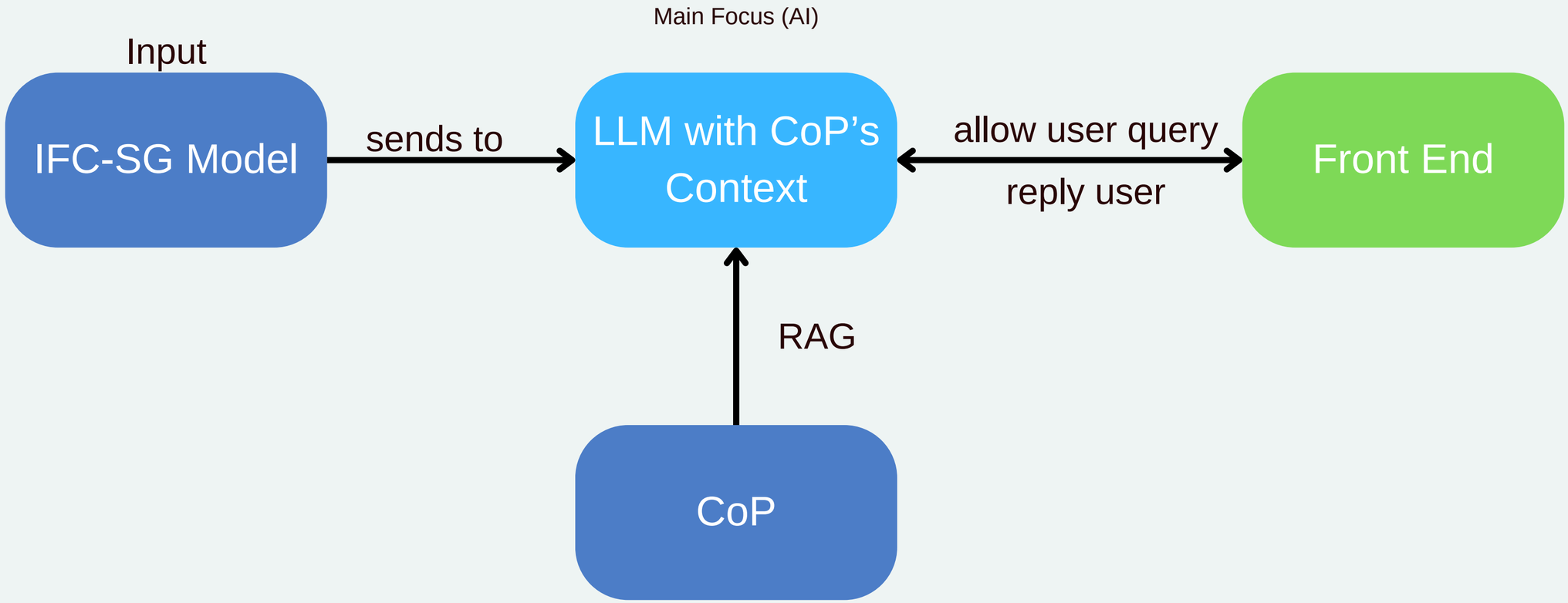
Significant effort required from the POs, diverting their focus from other critical tasks.

GOVTECH'S INTEREST:

To leverage AI for simplifying or automating all or parts of the review process.



THE AI MODEL WAS TRAINED TO LEARN FROM AGENCIES' CODE OF PRACTICE TO VALIDATE THE IFC-MODEL PROVIDED



THE AI MODEL WOULD ENABLE TIME SAVINGS FOR GOVERNMENT OFFICERS WORKING ON REGULATORY APPROVALS



Processing Officers (POs)

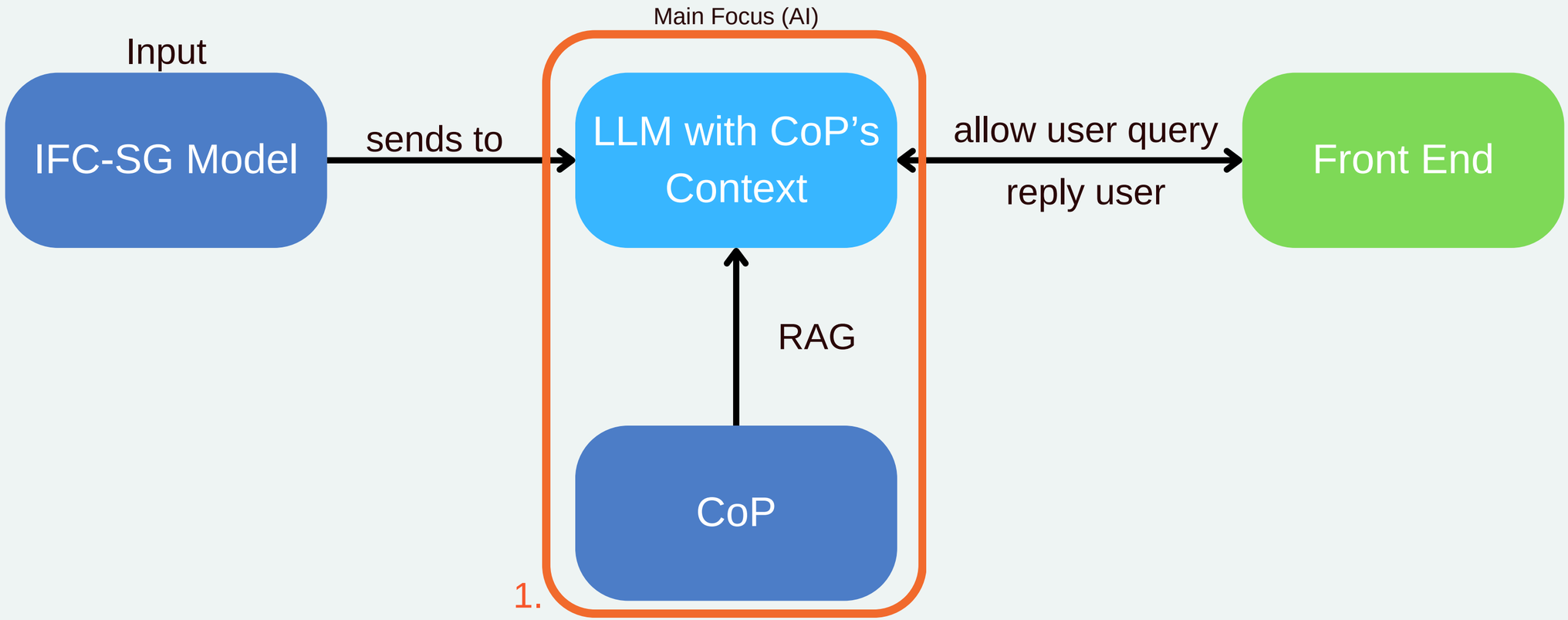
- Reduced effort required
- Minimal domain knowledge needed
- Time saved on training
- Decreased human error



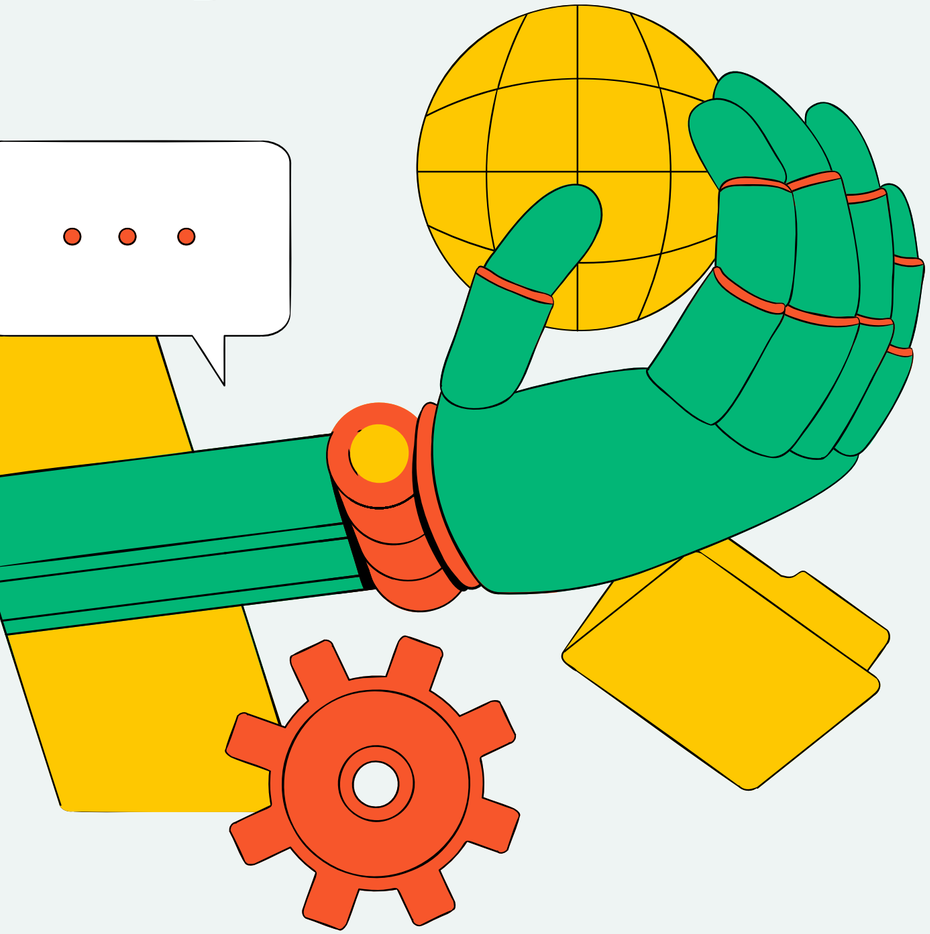
Government Agencies

- Reduced manpower requirements
- Improved time efficiency
- Cost Saving
- Enhanced accuracy and consistency

THE FOLLOWING SLIDES WILL DESCRIBE THE PROCESS TO DEVELOP THE AI MODEL TO UNDERSTAND THE CODE OF PRACTICE THROUGH NLP WORKFLOWS



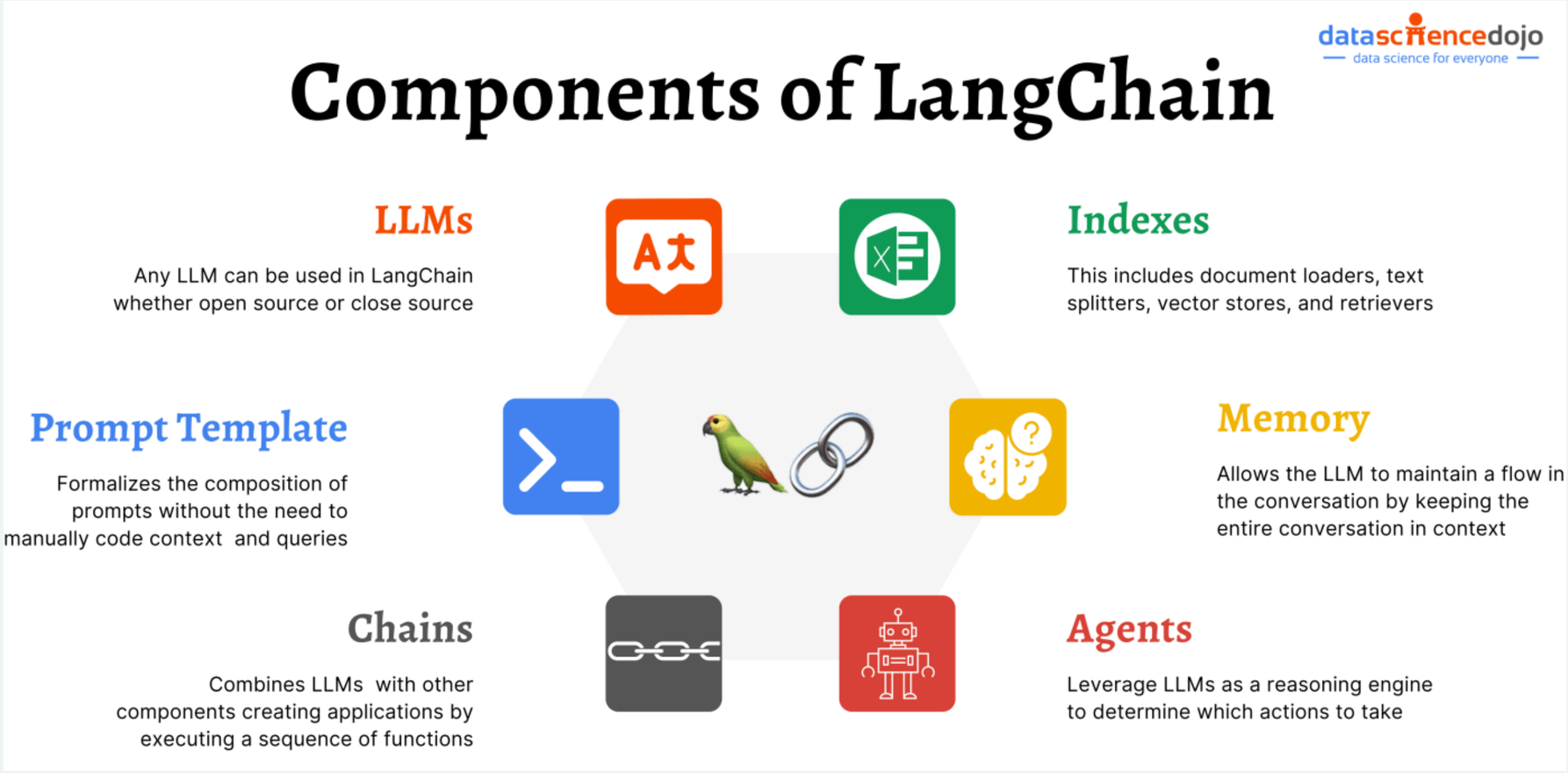
Retrieval–Augmented Generation (RAG) enhances LLM by integrating data on the fly



LLMs like GPT-4 were constrained by the static knowledge it was trained on, which could become outdated.

RAG enhances LLMs by integrating real-time data retrieval into the generation process, updating the model’s knowledge base on the fly.

LangChain is an open-source framework for developing applications powered by large language models (LLMs). 🦜🔗



It facilitates the process of querying the vector store and combining it with generative responses from OpenAI models.

E.g. RAG = Strategy, Langchain = Toolkit

OVERVIEW OF PIPELINE FOR RAG

1

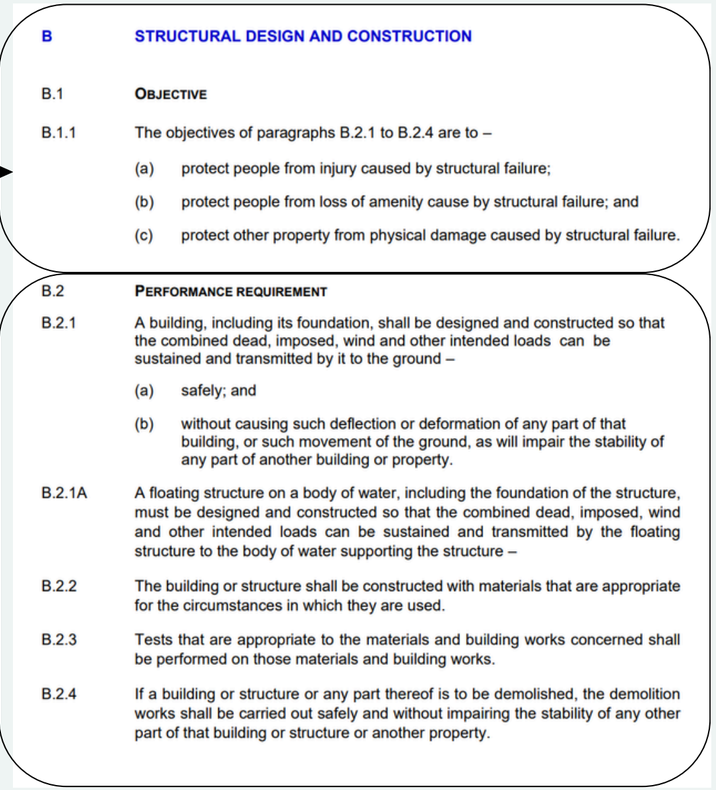
PDF

Code of Practice

split into

2

Chunk of Texts (from pdf)



B STRUCTURAL DESIGN AND CONSTRUCTION

B.1 OBJECTIVE

B.1.1 The objectives of paragraphs B.2.1 to B.2.4 are to –

(a) protect people from injury caused by structural failure;

(b) protect people from loss of amenity cause by structural failure; and

(c) protect other property from physical damage caused by structural failure.

B.2 PERFORMANCE REQUIREMENT

B.2.1 A building, including its foundation, shall be designed and constructed so that the combined dead, imposed, wind and other intended loads can be sustained and transmitted by it to the ground –

(a) safely; and

(b) without causing such deflection or deformation of any part of that building, or such movement of the ground, as will impair the stability of any part of another building or property.

B.2.1A A floating structure on a body of water, including the foundation of the structure, must be designed and constructed so that the combined dead, imposed, wind and other intended loads can be sustained and transmitted by the floating structure to the body of water supporting the structure –

B.2.2 The building or structure shall be constructed with materials that are appropriate for the circumstances in which they are used.

B.2.3 Tests that are appropriate to the materials and building works concerned shall be performed on those materials and building works.

B.2.4 If a building or structure or any part thereof is to be demolished, the demolition works shall be carried out safely and without impairing the stability of any other part of that building or structure or another property.

Using LangChain Tools

3

Embed into

4

Vector Database

1

When a user query

2

3

perform semantic search

4

Results

5

send into

Large Language Model (LLM)

6

return results to user

Legends:

How to convert documents to vector database

How to retrieve information based on the question out of the vector database

*Various AI models were explored and compared, refer to Annex A for for details

CHALLENGES FACED & STRATEGIES

1. SLOW REPLYING SPEED



Improve reply speed through different hardware (graphic card)

Theoretical Performance	
Pixel Rate:	118.1 GPixel/s
Texture Rate:	236.2 GTexel/s
FP16 (half):	15.11 TFLOPS (1:1)
FP32 (float):	15.11 TFLOPS
FP64 (double):	236.2 GFLOPS (1:64)

Red: RTX4060

Theoretical Performance	
Pixel Rate:	110.9 GPixel/s
Texture Rate:	277.3 GTexel/s
FP16 (half):	138.6 GFLOPS (1:64)
FP32 (float):	8.873 TFLOPS
FP64 (double):	277.3 GFLOPS (1:32)

Purple: GTX1080

The higher the FLOPS, the better as it allows for faster calculation

Using the IFC-SG Mapping Excel as inputs for training

	C	D	E	F	G
	Chapter	Regulatory	Regulatory	Regulatory	Detailed Regulatory Requirement
	Residential	7.1	General	N.A	The minimum width for accessible routes must be in accordance with clause 4.2.1.
	Accessibility	4.11.3	Detectable	N.A	(b) extend the full width of the stairs for a depth of 600 mm commencing 300 mm b
	Usage of	6.9.5	Family Car	N.A	Family car parking lots must have the minimum dimensions of 4800 mm long by 360
	Accessibility	4.12	Stairs for the	N.A	If provided, it must be designed to comply with the requirements of clause 4.11.1.1
	Accessibility	4.9.2	Lift Size	N.A	The minimum internal lift car dimension must be 1200 mm wide by 1400 mm deep.
	Accessibility	4.2.1	Width	N.A	The minimum width of the accessible routes, corridors and paths shall be: (a) as pr
	Provisions	2.3.5	Cinemas,	N.A	The wheelchair spaces, each measuring 900 mm wide by 1200 mm deep, must be lo
	Arriving At	3.6.1	Kerb Ramps	N.A	The width of a kerb ramp must be at least 900 mm
	Arriving At	3.5.1	Parking	N.A	Where vehicle parks are required to be provided, the number of accessible parking
	Accessibility	4.12	Stairs for the	N.A	(d) uniform risers of a maximum of 150 mm and treads of a minimum of 300 mm; ar
	Arriving At	3.4.2	Taxi Shelters	N.A	A taxi bay should, where possible, be provided at the level of approach for persons
	Arriving At	3.4.5	Taxi Shelters	N.A	Where there is a kerb between the access aisle and the vehicle pick-up space, it m
	Arriving At	3.2	Approach to	N.A	A passenger alighting and boarding point complying with clause 3.3 should, where
	Arriving At	3.4.2	Taxi Shelters	N.A	A taxi bay should, where possible, be provided at the level of approach for persons
	Arriving At	3.5.4	Location	N.A	The accessible route should not require wheelchair users to pass behind vehicles t
	Accessibility	4.2.5	Resting	N.A	(b) should be 50 m for footpaths and park connectors; and
	Accessibility	4.2.5	Resting	N.A	(b) should be 50 m for footpaths and park connectors; and

Compared to PDF, the Excel sheets did not have enough information, so the LLM reply seems not confident.

Experiment with alternate open-source Vector Database

Vector Database	Time taken to load the full PDF	Time taken to answer a query
Chroma	1min 24s	57s
FAISS	1min 10s	1min 8s

ChromaDB is the superior choice compared to other open-source options as it does not require cloud storage.

*Refer to Annex B for more research details

2. LLMS UNABLE TO RESPOND QUERIES BEYOND THEIR INPUT CONTEXT



The Use of Prompt Template

A prompt template guides the LLM on how to respond in a specific manner. It could aid the LLM in anwering questions beyond what it was provided for training (CoP).

Cody

```
prompt_template = """Use the following pieces of context to answer the question at the end.
        If you don't know the answer, please think rationally answer from your own knowledge base

        {context}

        Question: {question}
        """
```

In this case, the prompt template states that if it could not answer with question based on the content given, then answer from its own kowledge base.

Without the prompt template, LLMs are unable to answer.



Usage of Prompt Engineering to fine-tune replies from LLM

Prompt engineering refers to the process of designing and refining the inputs (prompts) given to a language model or other AI systems to achieve desired outputs.

Question: “I have a **corridor width** of 1000mm in **MRT Stations**, does it fufill the requirements?”

```
Follow-up Answer: Based on the provided context:

For existing buildings undergoing additions and alterations with exceptional circumstances and particular constraints, a corridor width of 1200 mm may be considered wi
th the provision of passing spaces (Clause 4.2.1(b) in Chapter 4).

However, your corridor width is 1000mm, which is less than the minimum required.

In addition, for MRT stations, Clause 4.2.1(a) mentions that the primary access routes must meet specific widths as prescribed in Table 3 (not provided here). The text
 does not specifically address MRT station requirements for corridors with a width of 1000mm.

Thus, according to the given context, it appears that your corridor width of 1000mm does not fulfill the general requirement for accessible routes and corridors.
```

Activate Windows

The answer given is incorrect, the minimum width should be 1800mm for MRT Stations.

TABLE 3
Accessible Route Widths

Building Type	Minimum width of accessible route
Residential developments, hotels, boarding houses, service apartments, worker dormitories, hostels, halls of residence or dormitories, etc. Schools, office buildings Factories, workshops, industrial buildings and offices/showroom areas in warehouses	1500 mm
Universities, colleges and institutions of learning Parks, open spaces and places of public resort Sports complexes and public swimming pools	
MRT stations, bus interchanges and passenger terminals	1800 mm
Shopping complexes and multi-purpose complexes Markets, hawker or food centres, restaurants and eating establishments Hospitals, health-care centres, nursing homes, welfare homes and homes for the aged	

Used Prompt Engineering to improve the results from LLM

Question: “Does an **accessible route** of 1000mm in MRT Stations fufill the minimum width of accessible route requirement?”

Llama 3.1

```
The minimum width of accessible routes for MRT stations is 1800 mm.
```

GPT-4o Mini

```
According to TABLE 3 in Chapter 4, the minimum width of accessible route for MRT stations is 1800 mm.
```

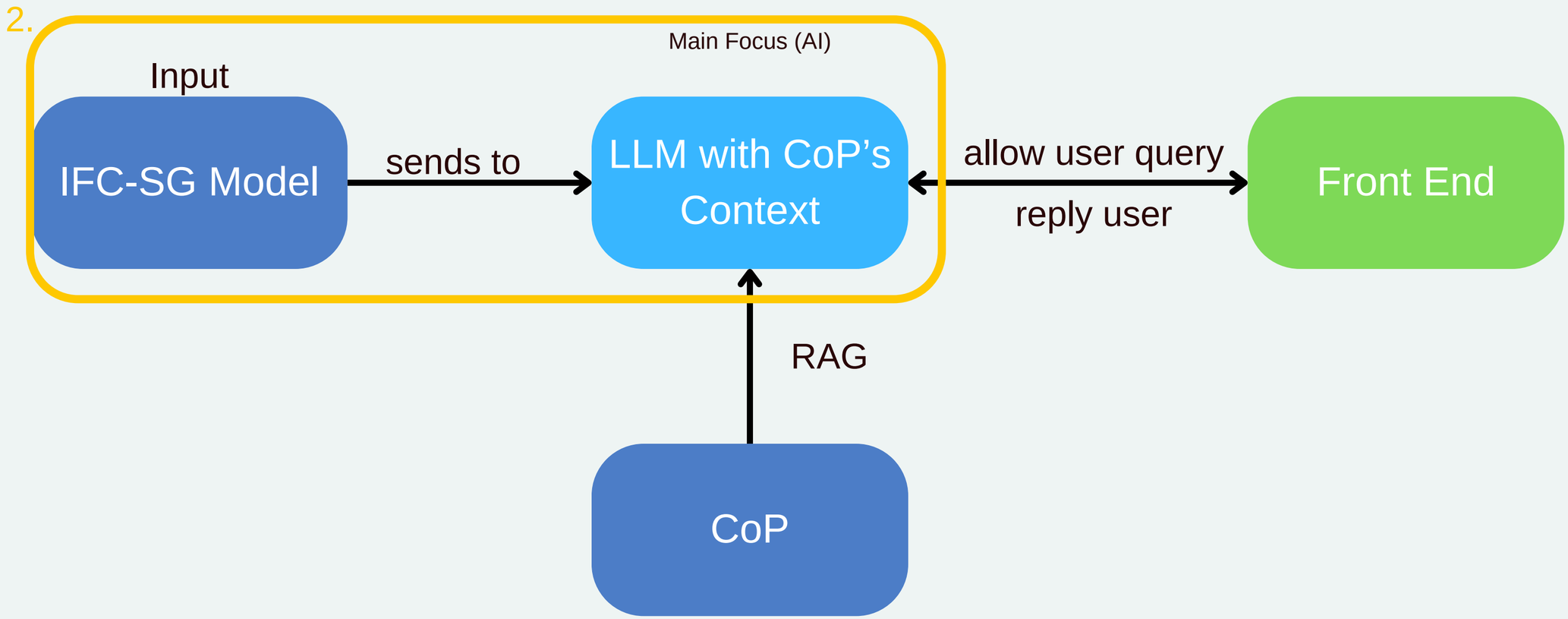
Overall Conclusion: Both of the models are able to answer **correctly**

This prompt works because “accessible route” was used instead of “corridor” in Table 3.

TABLE 3
Accessible Route Widths

Building Type	Minimum width of accessible route
Residential developments, hotels, boarding houses, service apartments, worker dormitories, hostels, halls of residence or dormitories, etc. Schools, office buildings Factories, workshops, industrial buildings and offices/showroom areas in warehouses	1500 mm
Universities, colleges and institutions of learning Parks, open spaces and places of public resort Sports complexes and public swimming pools MRT stations, bus interchanges and passenger terminals Shopping complexes and multi-purpose complexes Markets, hawker or food centres, restaurants and eating establishments Hospitals, health-care centres, nursing homes, welfare homes and homes for the aged	1800 mm

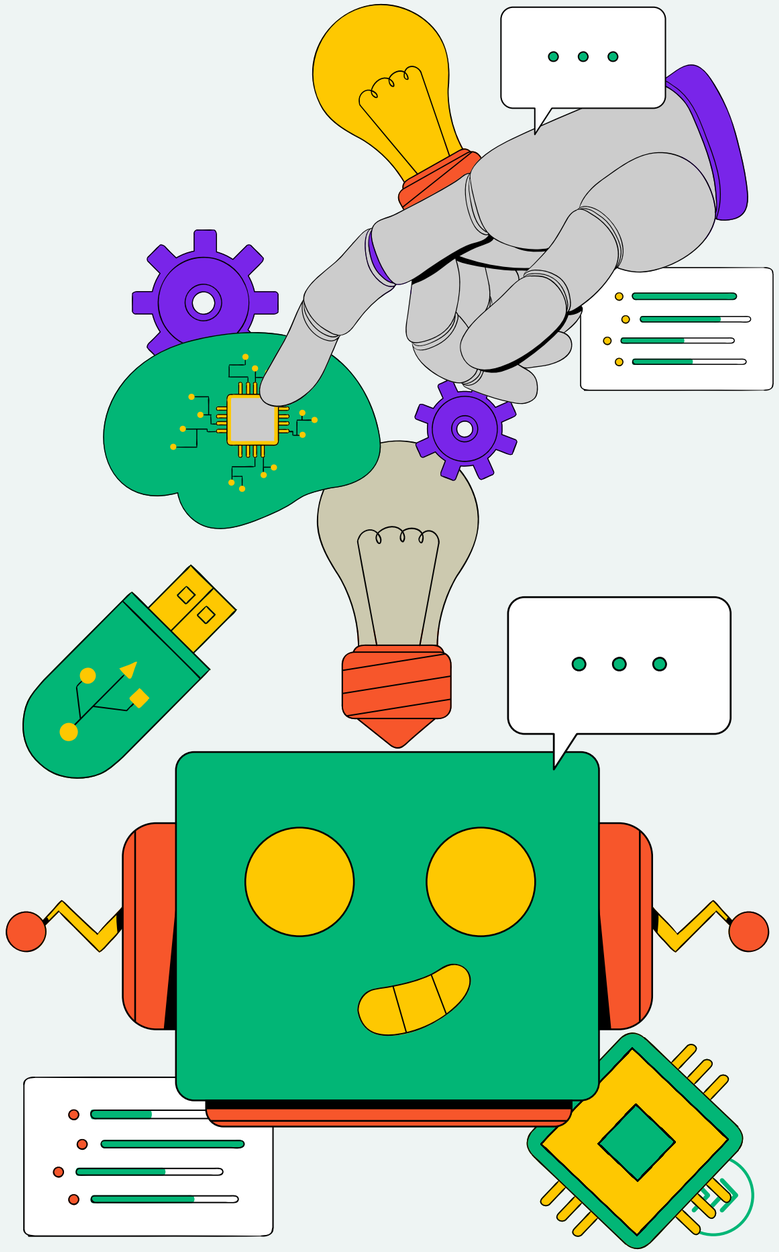
THE FOLLOWING SLIDES WILL DESCRIBE THE PROCESS TO DEVELOP THE AI MODEL TO UNDERSTAND THE IFC-SG MODELS THROUGH LLM



Enabling the LLM to interpret IFC-SG models

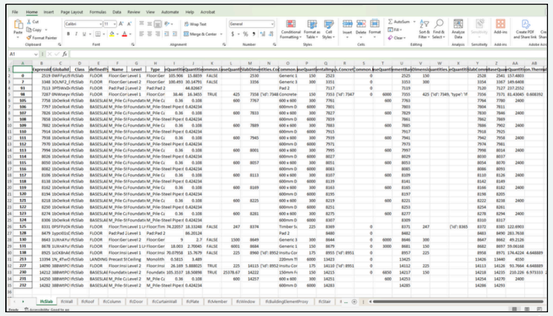
With current technologies, LLMs are not able to take in IFC-SG Models directly.

Since LLM cannot directly interpret IFC-SG models, strategies must be implemented to enable the LLM to understand the structure of these models.

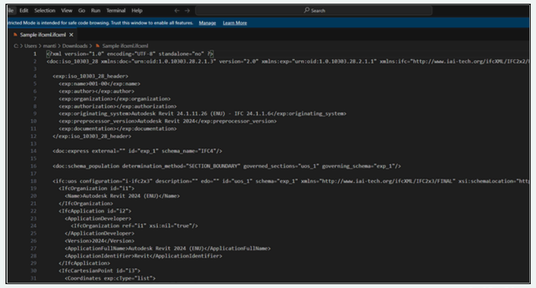


CHALLENGES FACED & STRATEGIES

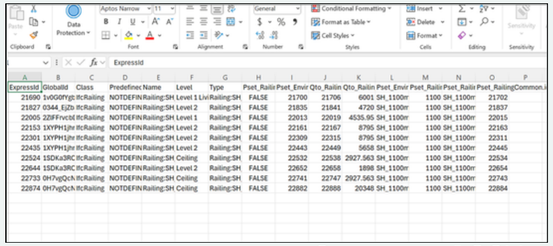
Experiment with different input format for LLM



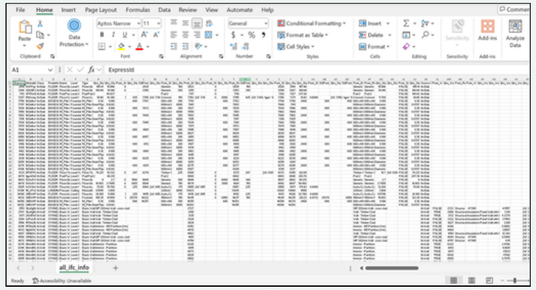
Excel



XML



Separate of CSVs



Consolidate of CSVs

As IFC is an open format, optimal results were achieved when information were converted into multiple CSV files.

Usage of Prompt Template

Before

Analyse and return the requested instances. Each result must contain the following properties (if available):
- GlobalID
- ExpressID

with few-shot

User Query: Return IfcWindow
Model reply:
- GlobalID = Win352
- ExpressID = 3526723

Before

Query: How many windows are there?

Answer: There are 16 instances of window.

with CoT

CoT: Identify the relevant sheet based on the query. Count the total number of rows in the relevant sheet(s), excluding the header row.

Query: How many windows are there?

Answer: Identified the IfcWindow sheet and there are 16 rows excluding header. Hence there are 16 instances of window.

Replies improved after implementing Few Shot and Chain-of-Thought (CoT) techniques.

*Refer to Annex C for more research details

EXPERIMENTING AI MODELS ON VARIOUS CLOUD PROVIDERS



Amazon Bedrock

- Provides access to pre-trained foundation models from leading AI providers.
- Simplifies building and scaling generative AI applications.



Open AI Assistant

- Allows us to build AI assistants. It can leverage models, tools, and knowledge to respond to user queries.
- The Assistants API currently supports three types of tools:
  - 1.File Search
  - 2.Code Interpreter
  - 3.Function Calling

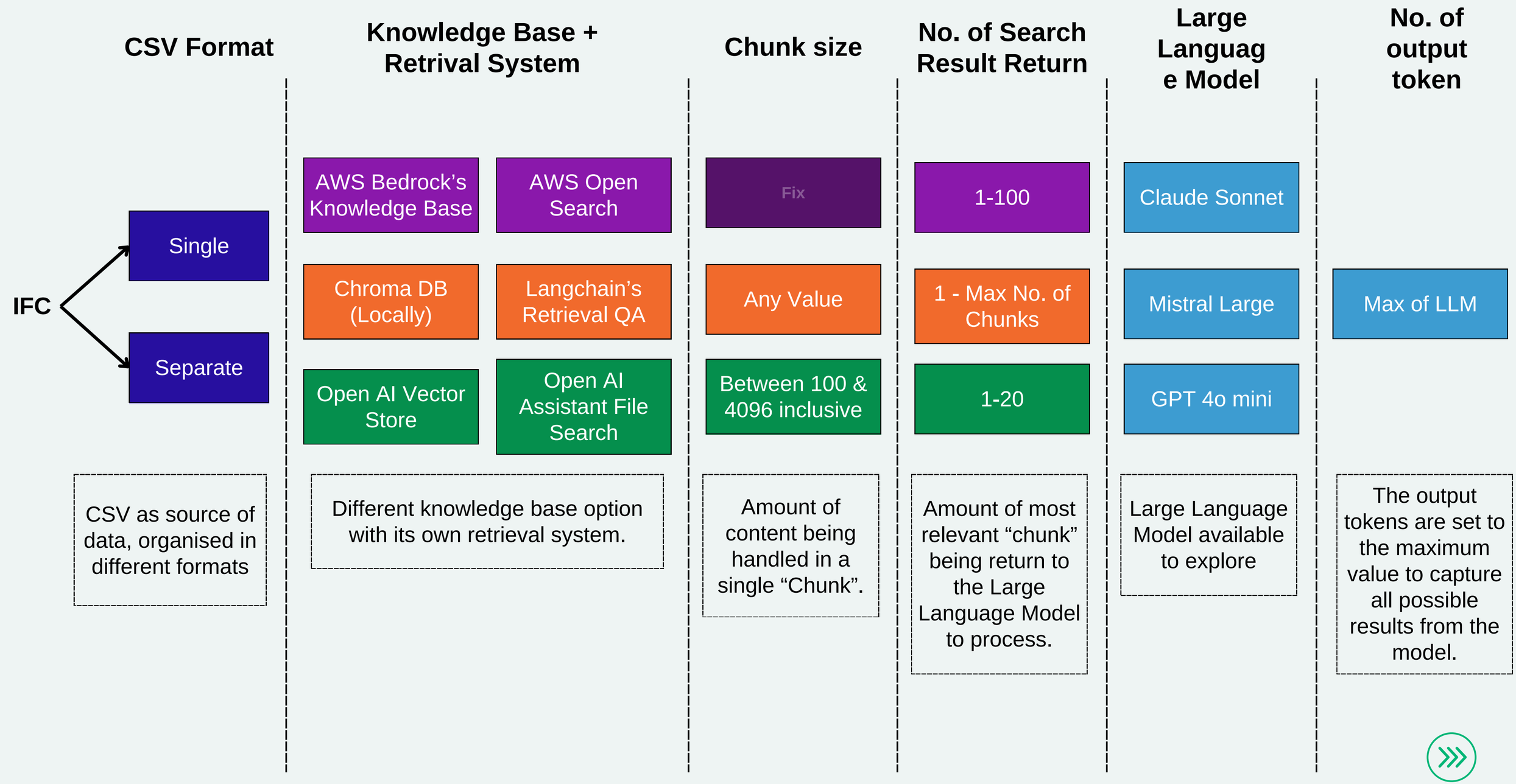


Azure AI

- Integrates AI models with external data sources to deliver context-aware and dynamic responses
- Works with other Azure AI services, such as Azure Machine Learning, Azure Bot Services, and Azure OpenAI.

*Refer to Annex D for more research details

EXHAUSTIVE TEST TO DERIVE BEST OUTCOME FOR INFORMATION SEARCHES THROUGH TABLE


CSV Format
Knowledge Base + Retrival System
Chunk size
No. of Search Result Return
Large Languag e Model
No. of output token
IFC
Single
Separate
AWS Bedrock's Knowledge Base
AWS Open Search
Chroma DB (Locally)
Langchain's Retrieval QA
Open AI Vector Store
Open AI Assistant File Search
Fix
Any Value
Between 100 & 4096 inclusive
1-100
1 - Max No. of Chunks
1-20
Claude Sonnet
Mistral Large
GPT 4o mini
Max of LLM
CSV as source of data, organised in different formats
Different knowledge base option with its own retrieval system.
Amount of content being handled in a single "Chunk".
Amount of most relevant "chunk" being return to the Large Language Model to process.
Large Language Model available to explore
The output tokens are set to the maximum value to capture all possible results from the model.

Evaluate the Data Completeness of the AI Model

Completeness Ratio

- A metric often used in various fields, including finance and data analysis, to measure how much of a dataset, project, or task has been completed relative to the total expected or required amount.
- Completeness Ratio = $\left(\frac{Total\ no.\ of\ Requirement\ met}{Total\ no.of\ requirement\ set}\right)$ x 100%

Evaluate the Data Completeness of the AI Model

Completeness Ratio

- Structured Evaluation
  - Checklist offers a clear framework for accessing model responses
- Identify Strengths and Weaknesses
  - Highlights specific areas where the model excels and where it requires improvement
- Complexity of Responses
  - Our project focuses on generating open-ended text that necessitates human evaluation.

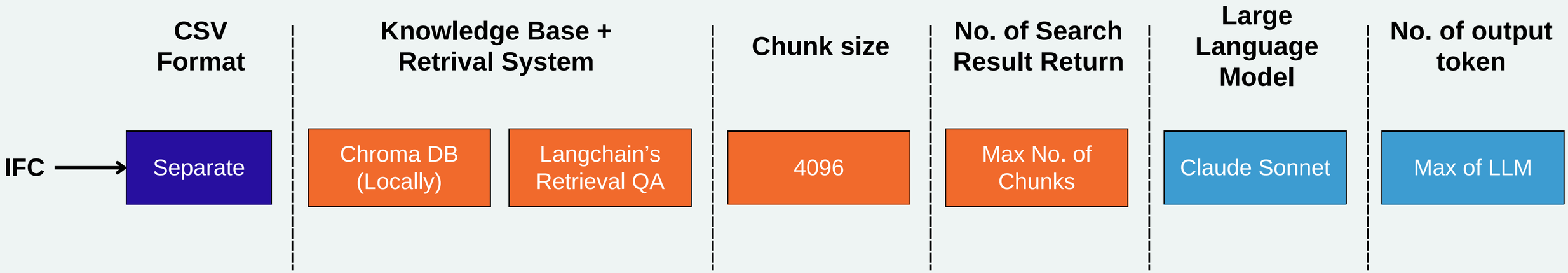
Evaluate the Data Completeness of the AI Model

Completeness Ratio

Requirements set:

- Are the returned GlobalID, ExpressID, Name of the object, and Dimensions accurate?
- Were all instances of the entity correctly retrieved?
- Does the response contain hallucinated information?
- Is the reply structured in the required format?
- Did it return any non-relevant entities?

SELECTION OF FINAL PIPELINE FOR EXHAUSTIVE TEST SUITE

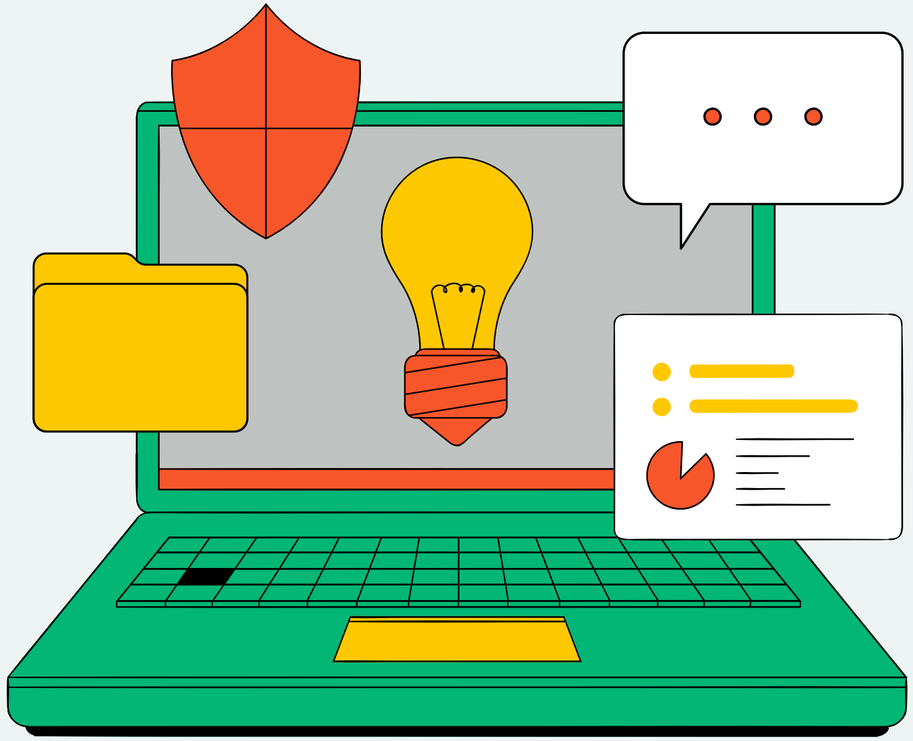


Completeness Ratio: **84.8%**

Open Component Front-end

Integrate RAG and IFC-SG knowledge into the LLM for processing.

Develop an IFC with an open component interface to allow users to interact and ask questions.



Libraries Utilized

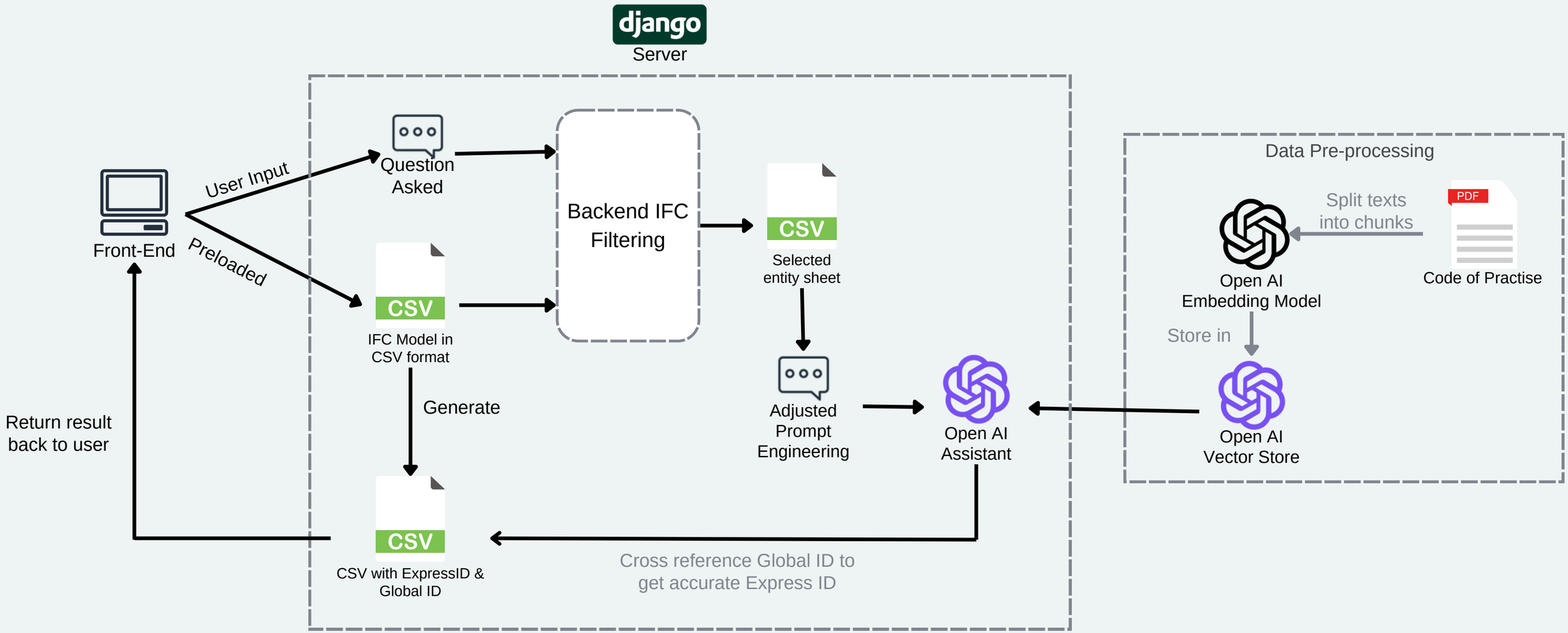
That Open Engine

- That Open Engine is an open-source 3D BIM engine designed for AECO software development.
- Supports JavaScript/TypeScript
- Adaptable for web-based and desktop applications.

Django

- Acts as the backend server
- Facilitates API calls to OpenAI
- No Installation of Application is required
- Lightweight for User

FINAL SYSTEM ARCHITECTURE



DEMONSTRATION

